

Chatterbox-Flash: Prior-Calibrated Block Diffusion for Streaming Zero-Shot TTS

Deokjin Seo*

Resemble AI

deokjin.seo@resemble.ai
ejrwlsls012@gmail.com

Gangin Park*

Seoul National University

ssonpull519@gmail.com

Kihyun Nam

KAIST

nk.h.mmai@kaist.ac.kr

Abstract

We present Chatterbox-Flash, a zero-shot text-to-speech model obtained by fine-tuning a pretrained autoregressive TTS decoder into a block-diffusion decoder, enabling parallel token generation within each block while retaining block-by-block streaming. We find that naively transferring mainstream block-diffusion decoding to discrete speech tokens degrades quality, as a long-tail token distribution biases parallel position selection toward a few high-frequency tokens. To mitigate this without architectural modification, we introduce two inference-time techniques: prior-calibrated scoring, which subtracts the block-level marginal token distribution, and an early-decoding schedule, which adaptively terminates iteration based on calibrated confidence. On standard zero-shot TTS benchmarks, Chatterbox-Flash attains high-fidelity synthesis comparable to strong autoregressive and non-autoregressive baselines, while supporting streaming inference with time-to-first-packet on par with streaming AR systems and substantially lower real-time factor. Code is available at <https://anonymous.4open.science/r/chatterbox-flash-B458>.

1 Introduction

Zero-shot text-to-speech (TTS), capable of synthesizing speech in unseen speakers’ voices from a short reference audio clip, has emerged as a central capability of modern speech synthesis, driven by large-scale multi-speaker training (Wang et al., 2023; Le et al., 2023; Borsos et al., 2023; Anastassiou et al.).

Recent zero-shot TTS models can be broadly categorized along two axes. Along the axis of generation order, autoregressive (AR) models generate tokens sequentially, conditioning each step on previous outputs (Wang et al., 2023; Du et al., 2024, 2025; Zhou et al., 2026), while non-autoregressive

(NAR) models produce all positions in parallel (Es-kimez et al., 2024; Chen et al., 2025; Wang et al., 2025). Along the axis of modeling space, they operate either on discrete tokens derived from neural audio codecs (Défossez et al., 2022; Zeghidour et al., 2021) or on continuous latent representations (Shen et al., 2024). Among these combinations, AR language models over discrete audio codecs have proven particularly effective, delivering strong speech quality while remaining amenable to streaming inference (Yang et al., 2026).

Despite these advantages, AR decoding incurs an inherent latency since tokens are produced sequentially and wall-clock time grows linearly with output length—a bottleneck that cannot be removed by engineering alone, motivating model-native parallel decoding. Among such approaches, diffusion language models (DLMs) (Austin et al., 2021; Lou et al., 2024; Sahoo et al., 2024) generate multiple tokens per step and have recently achieved throughput far above AR models with minimal quality loss, with the LLaDA family (Nie et al., 2026; Bie et al., 2025, 2026) and other extensions (Li et al., 2026; Ye et al., 2025a) scaling the paradigm to discrete large language models (dLLMs). Block Diffusion (Arriola et al., 2025b) adds block-causal training and inference compatible with streaming, later made practical at scale by Fast-dLLM (Wu et al., 2025b) and Fast-dLLM v2 (Wu et al., 2025a).

However, DLMs remain underexplored for speech, with existing efforts limited in scale (Ku et al., 2026), lacking native streaming (Zhu et al., 2026), or, in a similar block-wise paradigm, pairing masked diffusion with a separately-parameterized AR module that only passes a compact hidden state rather than tokens (Song et al., 2025). Moreover, we find that DLM decoding techniques do not transfer directly to discrete speech: codec sequences are heavily skewed toward a few dominant tokens—notably silence—that carry little context-dependent information (Liu et al., 2025; Sicherman and Adi,

* Equal contribution.

2023), and block-by-block decoding restricts position selection to a small local window (Shu et al., 2026) where ranking is fragile and not directly supervised (Asano et al., 2026).

In this work, we present Chatterbox-Flash, a zero-shot TTS model obtained by fine-tuning a pretrained autoregressive decoder into a discrete block-diffusion decoder, retaining the original architecture and replacing only the training objective with masked denoising. To address the problem with degraded quality, we introduce two inference-time techniques—*prior-calibrated scoring* and an *early-decoding schedule*—and integrate them with a block-causal streaming pipeline that delivers streaming inference at substantially lower latency than AR baselines.

Our contributions are as follows:

- **Streaming Block-diffusion TTS** To our knowledge, the first zero-shot TTS model to perform block diffusion decoding in the sense of (Arriola et al., 2025b) — i.e., a language model that directly generates discrete codec tokens under block-causal attention over the full prefix — with native block-by-block streaming.
- **Prior-Calibrated Scoring** An inference-time correction that suppresses the long-tail token bias in parallel position selection, requiring no architectural change or additional forward pass.
- **Early-Decoding Schedule** An adaptive termination rule that lowers the average number of denoising steps below the maximum budget based on calibrated confidence.
- **Empirical Validation** Chatterbox-Flash matches strong AR and NAR baselines in quality while being the only system in our comparison with native streaming support.

2 Method

2.1 Modeling

Architecture Our system extends Chatterbox-TTS (Resemble AI, 2025)¹, an open-source two-stage zero-shot TTS pipeline. Stage 1 is a Llama-style Transformer decoder (T3) that performs next-token prediction over a discrete speech token sequence $\mathbf{y} = (y_1, \dots, y_T)$ extracted at 25 Hz by a

¹<https://github.com/resemble-ai/chatterbox>

neural audio codec. The decoder is conditioned on

$$\mathbf{c} = [\mathbf{e}_s, \mathbf{x}_{\text{text}}, \mathbf{x}_{\text{speech}}], \quad (1)$$

which combines a global speaker embedding $\mathbf{e}_s$ obtained from a GE2E-trained voice encoder (Wan et al., 2020), the input text token sequence $\mathbf{x}_{\text{text}}$, and the prompt speech tokens $\mathbf{x}_{\text{speech}}$ extracted from the reference audio. The speech distribution is factorized autoregressively as

$$p(\mathbf{y} | \mathbf{c}) = \prod_{t=1}^T p(y_t | y_{<t}, \mathbf{c}). \quad (2)$$

Stage 2 is a flow-matching vocoder that converts the generated tokens to waveforms with chunk-wise streaming; it follows the CosyVoice2 (Du et al., 2024) architecture, distilled into a few-step flow-matching sampler with MeanFlow (Geng et al., 2025). We reuse this vocoder unchanged and focus our contribution on the Stage-1 decoder, so quality differences across decoding configurations are attributable to

Block Diffusion Following (Arriola et al., 2025b; Wu et al., 2025a), we apply discrete denoising diffusion block-by-block on the T3 decoder instead of over the full sequence. A length- T sequence x is partitioned into $B = \lceil T/D \rceil$ non-overlapping blocks $x^{(1)}, \dots, x^{(B)}$ of size D , giving

$$p(x) = \prod_{b=1}^B p(x^{(b)} | x^{(<b)}). \quad (3)$$

Each factor is modeled by a parallel masked predictor: positions in $x^{(b)}$ are randomly replaced with $[M]$ to form $x_t^{(b)}$, and the predictor recovers the masked tokens in parallel from $x^{(<b)}$ and $x_t^{(b)}$. The left-to-right inter-block factorization naturally supports block-wise streaming generation.

2.2 Training

Packed Input and Hybrid Attention The packed input $[\mathbf{c}, x_t]$ uses a hybrid attention scheme over B speech blocks (Figure 3): causal over the conditioning $\mathbf{c}$, bidirectional within each speech block, and causal across blocks, so that $x_t^{(b)}$ attends only to $\mathbf{c}$ and $x^{(<b)}$. Unlike Fast-dLLM v2 (Wu et al., 2025a), which is block-causal throughout, we keep $\mathbf{c}$ causal to preserve the pretrained backbone’s embedding space and apply block-diffusion attention only to the speech part, maintaining the monotonic text-to-speech alignment. We implement this with custom attention kernels (Appendix A).

Complementary Masking At each training step we sample $t \sim \mathcal{U}(\epsilon, 1 - \epsilon)$, derive the per-token mask probability from a fixed noise schedule, and draw a binary mask $m \in \{0, 1\}^T$ over the speech positions to form x_t ($x_{t,i} = y_i$ if $m_i = 0$, else $[M]$). Following Fast-dLLM v2 (Wu et al., 2025a), we add the complementary view $\bar{m} = 1 - m$ as a second sample in the same batch, so every position is supervised under both masked and unmasked contexts.

Token-Shift Denoising Loss We adopt a next-token prediction parameterization following Fast-dLLM v2 (Wu et al., 2025a): a masked position i is predicted from the hidden state at position $i - 1$ rather than from the mask token itself. This shifted-label form preserves the backbone’s autoregressive interface while still allowing bidirectional context within each block. The per-token cross-entropy loss at a masked position i is

$$\ell_i = -\log p_\theta(y_i \mid \mathbf{c}, x^{(<b(i))}, x_t^{(b(i))}), \quad (4)$$

where y_i is the clean target and $b(i)$ is the block containing i ; the prediction is conditioned on $\mathbf{c}$, the clean preceding blocks $x^{(<b(i))}$, and the noised current block $x_t^{(b(i))}$. With $\mathcal{M}_b = \{i \mid x_{t,i} = [M]\}$ the masked positions in block b (excluding padding), the sample-level loss averages per-token losses within each block and then over blocks,

$$\mathcal{L}_{\text{denoise}} = \frac{1}{B} \sum_{b=1}^B \frac{1}{|\mathcal{M}_b|} \sum_{i \in \mathcal{M}_b} \ell_i. \quad (5)$$

2.3 Inference

2.3.1 Block-Autoregressive Decoding

Block-diffusion inference commits the sequence one block at a time in left-to-right order, with masked positions in the current block unmasked in parallel. Already-committed blocks act as clean context, and their key-value caches are appended sequentially to provide the inter-block context for subsequent blocks. Since the conditioning prefix is encoded causally and never attends to speech tokens, its key-value cache depends only on $\mathbf{c}$ and is computed once at the start of inference and reused across every block. At each step, the forward pass therefore operates only on the current block, accessing the prefix and previously committed blocks through their appended caches.

At each step within a block, the forward pass produces a predictive distribution $p_i^{(k)}$ and a predicted token $\hat{x}_i^{(k)} = \arg \max_v p_i^{(k)}(v)$ at every

masked position i . Two decisions must then be made: *which* positions to commit at this step (Section 2.3.2) and *how many* positions to commit (Section 2.3.3). When classifier-free guidance is used (Section 2.3.4), $p_i^{(k)}$ refers to the conditional branch alone, while a separate guidance combination determines $\hat{x}_i^{(k)}$.

2.3.2 Prior-Calibrated Scoring

Block-level commitment can induce *boundary-induced context truncation* (BICT) (Shu et al., 2026): once a few positions in a block are committed to incorrect tokens, subsequent blocks decode on top of corrupted context. This is particularly pronounced for discrete speech codecs, where strong local acoustic dependencies between adjacent frames coexist with a small set of *dominant tokens*—such as silence or low-energy frames—that occupy a disproportionate share of the marginal distribution (Liu et al., 2025; Asano et al., 2026); mistakenly committing such a token near a block boundary breaks acoustic continuity.

To address this, we propose *prior-calibrated scoring*, which assigns each masked position an ordering score for parallel unmasking. The common choice is the model’s per-position confidence $p_i^{(k)}(\hat{x}_i^{(k)})$, but for discrete speech codecs this tends to assign large values to dominant tokens regardless of context, causing them to be unmasked preferentially. To separate this marginal bias from the contextual prediction, we use a pointwise mutual information (PMI) score,

$$s_i^{(k)} = \log p_i^{(k)}(\hat{x}_i^{(k)}) - \log \bar{p}(\hat{x}_i^{(k)}), \quad (6)$$

where $\bar{p}$ is a reference distribution measuring the marginal probability of producing $\hat{x}_i^{(k)}$ irrespective of local context. The first term is the model’s log-confidence at position i , and the second subtracts the marginal share of the same token under $\bar{p}$, so that $s_i^{(k)}$ measures how specifically the predicted token is licensed by local context.

A natural choice for $\bar{p}$ is the in-block marginal averaged over the current block’s predictive distributions, but this prior is itself shaped by $\mathbf{c}$, making the score partially self-referential. We instead use the *unconditional block prior*, computed once from a single forward pass on all-masked sequence $[M]^D$ with conditioning embeddings zeroed,

$$\bar{p}(v) = \frac{1}{D} \sum_{j=1}^D p_\theta(v \mid [M]^D, \mathbf{c} = \mathbf{0})_j. \quad (7)$$

Since $\bar{p}$ depends only on (D, θ) , it is cached for the lifetime of the model.

2.3.3 Unmasking Schedule

Beyond which positions to unmask, the number of positions unmasked at each step must also be chosen. Committing too many positions in a single step risks introducing incorrect tokens as misleading context, while committing too few requires nearly all K steps and increases inference cost.

Time-Shifted Schedule LaViDa (Li et al., 2026) introduced a time-shifted (TS) schedule that biases unmasking away from uniform, also adopted in OmniVoice (Zhu et al., 2026) as its decoding rule: at each step, the TS schedule determines *how many* positions to unmask, and the model commits the corresponding number of top-confidence positions. We build on the same TS schedule for time allocation, replacing only the position-selection criterion with prior-calibrated scoring (Section 2.3.2). The cumulative fraction of unmasked tokens at step k follows

$$r_k = \frac{\tau \cdot (k/K)}{1 + (\tau - 1) \cdot (k/K)}, \quad (8)$$

with $r_0 = 0$, total steps K , and shift parameter τ ; the fraction of newly unmasked tokens at step k is $f_k = r_k - r_{k-1}$. The optimal τ differs across configurations such as K and the target token distribution, and we explored different τ values in our experiments.

Early Decoding We further adapt the unmasking fraction at each step based on the prior-calibrated scores $s_i^{(k)}$ of Equation (6). At step k , positions whose score exceeds a threshold θ_k are unmasked,

$$\theta_k = \text{Quantile}(\{s_i^{(k)}\}_{i \in \mathcal{M}}, q_k), \quad (9)$$

$$q_k = \max\left(0, 1 - \alpha \cdot \frac{k+1}{K}\right), \quad (10)$$

where $\mathcal{M}$ is the set of masked positions in the current block and $\alpha \in [0, 1]$ controls how quickly the threshold relaxes. At early steps, q_k is close to one and only a few positions in the upper region are unmasked; as k increases, q_k decreases and more positions become eligible. Denoting the resulting fraction by $g_k = q_{k-1} - q_k$, we combine the two schedules via $\max(f_k, g_k)$, so that the TS schedule provides per-step commits while early decoding adds further commits when calibrated confidence is high. Decoding terminates when all positions in the block have been unmasked or when K steps

have been reached; under large α , the average step count drops below K . Following OmniVoice (Zhu et al., 2026), we additionally support sampling temperatures at both the token and position levels, the latter realized as Gumbel-perturbed selection over the prior-calibrated scores with temperature $\beta = 5$; their effect is studied in Section 3.6.

2.3.4 Classifier-Free Guidance

We combine classifier-free guidance (CFG) (Ho and Salimans, 2022) with prior-calibrated decoding by running both a conditional and an unconditional forward at each step, the latter with the conditioning embeddings replaced by zero vectors, yielding logits ℓ_i^c and ℓ_i^u . Token sampling uses the standard combination $\ell_i = (1+w)\ell_i^c - w\ell_i^u$ to determine the predicted token $\hat{x}_i^{(k)} = \arg \max_v \text{softmax}(\ell_i)_v$, while the prior-calibrated score is evaluated on the conditional branch alone (i.e., $p_i^{(k)} = \text{softmax}(\ell_i^c)$ in Equation (6)). This decoupling—CFG-guided sampling for the committed token, conditional-only PMI for position ranking—keeps the ranking insensitive to w while still routing the CFG-guided token through Equation (6). We use $w = 1.0$ by default; a full sweep of w and an alternative mode combining s_i^c and s_i^u are provided in Section C.2.

3 Experiments

3.1 Experimental Setup

Evaluation Benchmarks We evaluate on two publicly available English zero-shot TTS benchmarks. LibriSpeech-PC test-clean is a zero-shot voice cloning benchmark built on the test-clean split of LibriSpeech-PC (Meister et al., 2023), and Seed-TTS test-en is the English evaluation set introduced by Seed-TTS (Anastassiou et al.).

Metrics We report SIM-o (cosine similarity between WavLM-ECAPA-TDNN (Desplanques et al., 2020; Chen et al., 2022) speaker embeddings of generated vs. reference speech), WER (HuBERT (Hsu et al., 2021) ASR transcription on LibriSpeech-PC, Whisper-large-v3 (Radford et al., 2022) on Seed-TTS test-en, against the input text), and UTMOS (Saeki et al., 2022) for naturalness. Metric configuration and baseline numbers are taken from OmniVoice (Zhu et al., 2026).

Implementation Details The model replaces the T3 decoder of Chatterbox-TTS with the block-diffusion architecture described in Section 2, trained with the hybrid attention mask and the

Model	#Params	Steps	LibriSpeech-PC test-clean			Seed-TTS test-en		
			SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
Ground-truth	–	–	0.690	1.87	4.10	0.734	2.14	3.52
<i>Autoregressive Models</i>								
IndexTTS2 (Zhou et al., 2026) ‡	1.7B	–	0.700	2.35	4.06	0.706	2.33	3.65
CosyVoice3 (Du et al., 2025) ‡	1.1B	–	0.694	1.59	4.28	0.696	2.17	3.96
VoxCPM (Zhou et al., 2025) ‡	0.7B	–	0.717	1.74	4.18	0.731	1.92	3.77
Qwen3-TTS (Hu et al., 2026) ‡	1.1B	–	0.704	1.60	4.41	0.708	1.54	4.16
Chatterbox (Resemble AI, 2025) ‡	0.5B	–	0.707	1.99	4.29	0.685	2.20	4.10
<i>Non-Autoregressive Models</i>								
F5-TTS (Chen et al., 2025) ‡	0.4B	–	0.655	1.89	3.89	0.664	1.85	3.72
ZipVoice (Zhu et al., 2025) ‡	0.1B	–	0.668	1.64	3.98	0.697	1.70	3.82
MaskGCT (Wang et al., 2025) ‡	2.2B	–	0.691	2.26	3.91	0.713	2.88	3.55
OmniVoice-Emilia (Zhu et al., 2026) ‡	0.8B	–	0.697	1.57	4.23	0.717	1.72	3.88
OmniVoice (Zhu et al., 2026) ‡	0.8B	–	0.729	1.30	4.28	0.741	1.60	3.91
<i>Block-Autoregressive Models (Chatterbox-Flash, Ours)</i>								
w/ Fast-dLLM v2 decoding ‡	0.5B	10	0.656	15.36	4.14	0.646	14.49	4.00
w/ TS schedule	0.5B	8	0.714	1.69	4.29	0.703	1.97	4.09
w/ TS schedule + ED ($\alpha = 0.5$)	0.5B	5.55	0.703	1.83	4.28	0.693	2.20	4.08
w/ PMI ($\alpha = 0$, ours)	0.5B	8	0.717	1.67	4.29	0.704	1.96	4.09
w/ PMI + ED ($\alpha = 0.5$, ours)	0.5B	6.4	0.713	1.67	4.28	0.704	2.04	4.08

Table 1: Objective evaluation on zero-shot TTS benchmarks under **offline** decoding. Best results per AR / NAR / Block-AR group are in **bold**. Rows marked ‡ are reproduced from OmniVoice (Zhu et al., 2026); ours use the same test sets and metrics (Section 3.1), so all columns are comparable. **Steps** is the average denoising steps per block (Chatterbox-Flash only). Block-AR variants share one configuration ($D=16$, $\tau=0.5$, $w=1.0$, $T=0.2$, $\beta=5$), differing only in decoding: Fast-dLLM v2 ($K=10$, threshold 0.3, top- p 0.95, batch 16), TS schedule ($K=8$), PMI ($K=8$), and both with early decoding (+ED, $\alpha=0.5$).

token-shift loss. We initialize from a pretrained Chatterbox-TTS checkpoint and continue training with AdamW using a cosine learning rate schedule (peak 10^{-5} , 10% warmup) with an effective batch size of 440 in bfloat16 precision. The model is trained with a block size of $D = 32$. Text inputs are normalized by a custom preprocessor that converts numbers, dates, times, and other non-standard tokens into their spoken forms before tokenization. At inference time, our canonical configuration is block size $D = 16$, number of denoising steps $K = 8$, TS schedule parameter $\tau = 0.5$, CFG scale $w = 1.0$, sampling temperature 0.2, and position temperature $\beta = 5$. We report two main settings differing only in the early-decoding parameter α : a quality-strongest setting at $\alpha = 0$ (no early decoding) and an efficiency-oriented setting at $\alpha = 0.5$. All experiments are conducted on NVIDIA H100 GPUs, and inference uses attention kernels and paged key-value cache management from FlashInfer (Ye et al., 2025b); further implementation details are provided in Section E.

3.2 Main Results

Table 1 reports zero-shot TTS performance against recent state-of-the-art autoregressive (AR) and non-autoregressive (NAR) baselines, under the canonical configuration without early decoding ($\alpha = 0$, $K = 8$).

Against Chatterbox, the AR backbone we build on, the AR-to-block-diffusion conversion improves both SIM-o and WER on the two benchmarks while leaving UTMOS unchanged, unlocking parallel decoding at preserved or improved perceptual quality. Within the NAR group, Chatterbox-Flash trails only OmniVoice on LibriSpeech-PC SIM-o and WER despite training on 70k hours of English versus OmniVoice’s 581k hours of multilingual data, attains the best NAR UTMOS on LibriSpeech-PC, and surpasses F5-TTS, MaskGCT, and OmniVoice-Emilia on WER. It is also the only NAR system here with native streaming support (Section G).

The *Block-Autoregressive Models* group compares five decoding configurations under this canonical setting. Fast-dLLM v2 decoding transfers poorly to discrete speech codecs (WER > 14 on

Mode	Method	LibriSpeech-PC test-clean			Seed-TTS test-en		
		SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
Offline	TS schedule ($\alpha=0$)	0.714	1.69	4.29	0.703	1.97	4.09
	TS schedule ($\alpha=0.5$)	0.703	1.83	4.28	0.693	2.20	4.08
	PMI ($\alpha=0$)	0.717	1.67	4.29	0.704	1.96	4.09
	PMI ($\alpha=0.5$)	0.713	1.67	4.28	0.704	2.04	4.08
Streaming	TS schedule ($\alpha=0$)	0.703	1.66	4.19	0.687	1.79	3.99
	TS schedule ($\alpha=0.5$)	0.702	1.76	4.19	0.689	1.75	3.99
	PMI ($\alpha=0$)	0.703	1.70	4.20	0.690	1.73	3.99
	PMI ($\alpha=0.5$)	0.702	1.80	4.20	0.690	1.96	4.00

Table 2: Offline vs streaming quality on the two standard benchmarks under the canonical configuration. Best per column within each mode in **bold**. Offline rows match the Block-AR group of Table 1.

Mode	Method	Overall WER $\downarrow$		Pronunciation $\downarrow$		Paraling. $\downarrow$		Foreign $\downarrow$	
		Off	Str	Off	Str	Off	Str	Off	Str
TS	$\alpha=0$	37.97	36.25	73.02	71.35	27.69	23.36	17.59	18.44
	$\alpha=0.5$	35.75	39.53	74.53	85.88	19.51	19.98	18.06	18.54
PMI	$\alpha=0$	34.86	35.10	71.04	72.32	19.55	19.60	18.52	18.05
	$\alpha=0.5$	34.79	35.68	70.28	72.36	19.61	20.32	18.90	18.93

Table 3: Per-category WER on EmergentTTS-Eval (Manku et al., 2026), offline (Off) vs. streaming (Str). Best value per column within the PMI/TS $\alpha=0.5$ streaming setting in **bold**; **red** marks the TS baseline’s degradation under streaming. Model-as-a-judge MOS is stable across all rows (3.50–3.55) and omitted. *Pronunciation*: rare or non-standard pronunciation; *Paraling.*: paralinguistic cues; *Foreign*: foreign words.

both benchmarks) despite a carefully tuned threshold: its fixed confidence bar fires first on the dominant low-information tokens, locking in a corrupted context for the rest of the block—the boundary-induced truncation that motivates prior calibration (Section 2.3.2). The TS schedule alone (top-confidence selection, as in OmniVoice) and PMI ($\alpha = 0$) achieve statistically comparable quality, so PMI’s decisive contribution is not raw quality but the calibration of its scores, which serves as a reliable thresholding signal for early decoding. Isolating this from the scheduling rule, PMI+ED attains lower WER than TS+ED at $\alpha = 0.5$ (1.67 vs. 1.83 on LibriSpeech-PC, 2.04 vs. 2.20 on Seed-TTS) at a comparable step budget, confirming that the calibrated signal, not the schedule alone, drives the quality-preserving step reduction. PMI+ED matches the no-ED TS baseline’s quality while cutting the average step count from 8 to ~ 6.3 /block, a $\sim 20\%$ reduction.

3.3 Offline vs Streaming

The results above are offline. Under our block-causal streaming pipeline (Section D), the two decoding methods stay close and the offline-to-streaming shift is small (Table 2), so quality holds under the streaming constraint. These benchmarks

saturate near WER 1.7–2.0, however, leaving little room to separate the methods, which motivates the harder evaluation that follows.

3.4 Challenging TTS Scenarios

The standard benchmarks are dominated by read speech, where the model’s confidence and the marginal token distribution rarely disagree. Prior calibration is built for harder cases, so we evaluate on EmergentTTS-Eval (Manku et al., 2026), which targets complex prosodic, expressive, and linguistically difficult utterances (Table 3). At the overall level the asymmetry is stark: moving from offline to streaming raises the TS baseline’s WER by 3.78 (35.75 $\rightarrow$ 39.53) but PMI’s by only 0.89 (34.79 $\rightarrow$ 35.68).

The gap is driven by *pronunciation*, the hardest category, where the TS baseline with early decoding jumps to 85.88 under streaming while PMI stays around 70–72; because this category carries the most word errors, its collapse drives the overall degradation. The other categories show where calibration does not help: *foreign words* depend on lexical coverage rather than position selection, and *paralinguistics* is capped near 19–20 by the backbone’s limited handling of non-lexical vocalizations. PMI’s advantage is thus real but category-

Metric	ElevenLabs v3	Chatterbox-Flash
NMOS mean $\uparrow$	4.04 ± 0.26	3.91 ± 0.23
% ≤ 2 $\downarrow$	12.9	8.6
% ≥ 4 $\uparrow$	80.0	67.1
SMOS mean $\uparrow$	3.50 ± 0.35	4.56 ± 0.18

Table 4: Human evaluation on 10 utterances from Seed-TTS test-en, 7 ratings each (70 per system), on 5-point Likert scales (NMOS: naturalness; SMOS: speaker similarity). Means are shown with 95% confidence intervals. Chatterbox-Flash is evaluated in its default offline configuration ($D=16$, $\alpha=0.5$).

specific—it recovers intelligibility where dominant-token bias corrupts position selection, and is neutral where the bottleneck lies elsewhere.

3.5 Human Evaluation

We conducted a side-by-side evaluation against ElevenLabs v3, a frontier commercial zero-shot TTS system, collecting naturalness (NMOS) and speaker-similarity (SMOS) ratings on 5-point Likert scales over 10 Seed-TTS utterances, 7 ratings each (70 per system). On NMOS (Table 4), Chatterbox-Flash and ElevenLabs v3 attain comparable means (3.91 vs. 4.04), and the two confidence intervals overlap substantially (± 0.23 vs. ± 0.26), so mean naturalness is statistically indistinguishable. The difference instead shows in the tails: Chatterbox-Flash produces fewer MOS- ≤ 2 ratings (8.6% vs. 12.9%), indicating fewer catastrophic failures even when the means match. On SMOS the means favor Chatterbox-Flash (4.56 vs. 3.50) and the intervals do not overlap (± 0.18 vs. ± 0.35), so the speaker-similarity advantage is a reliable effect rather than sampling noise.

3.6 Ablation Studies

We ablate two inference-time hyperparameters affecting the quality–compute trade-off at the canonical configuration: the block size D and the denoising step budget K in combination with the early-decoding parameter α .

Block Size Figure 2(a) sweeps the inference block size $D \in \{8, 16, 24, 32\}$ with the model trained at $D = 32$. SIM-o and UTMOS are essentially flat across the range, and WER stays within noise up to $D = 16$ before degrading sharply at $D \geq 24$. This degradation reflects the difficulty of parallel unmasking on large blocks: these configurations require committing more positions per

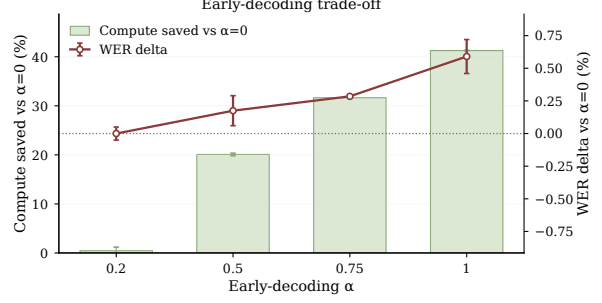


Figure 1: Early-decoding trade-off at $D = 16$, $K = 10$, averaged across LibriSpeech-PC and Seed-TTS test-en. Bars: number of computing steps saved relative to $\alpha = 0$. Line: WER delta versus $\alpha = 0$, with error bars spanning the two benchmarks. At $\alpha = 0.5$, $\sim 20\%$ steps are saved at negligible WER cost; further savings up to $\sim 41\%$ at $\alpha = 1$ incur a $\sim +0.6$ WER increase.

step than the model can confidently rank, even with prior calibration.

Step Budget Figure 2(b) characterizes the effect of step size K in fixed-step decoding at $D = 16$. $K \leq 2$ cannot recover the speech sequence, but quality stabilizes rapidly and plateaus by $K \geq 6$. We adopt $K = 8$ as the default, providing modest headroom over the plateau for adaptive early termination. The stars mark adaptive early decoding at $\alpha = 1$, which reaches the same plateau-level WER at ~ 4.6 average steps without exceeding the fixed-step plateau.

Early Decoding Figure 1 sweeps the early-decoding parameter $\alpha \in \{0.2, 0.5, 0.75, 1.0\}$ at $D = 16$, $K = 10$. At $\alpha = 0.2$ early decoding is almost inactive, but from $\alpha = 0.5$ onward it meaningfully tightens the step budget while the WER cost grows only mildly, staying within $+0.6$ of the no-ED baseline even at $\alpha = 1.0$. We adopt $\alpha = 0.5$ as the canonical efficiency-oriented setting ($\sim 20\%$ compute reduction at negligible quality cost); $\alpha = 0.75$ – 1.0 offer further savings for latency-critical deployments.

Further Ablations Sweeps of the CFG scale w , sampling temperature T , and position temperature β , together with a head-to-head comparison of the TS schedule baseline against PMI across step budgets $K \in \{2, 5, 8\}$, are provided in Section C. The comparison shows that PMI and the TS schedule baseline track each other closely across the step range, confirming that the choice of decoding method contributes only marginally to quality at any compute budget—PMI’s advantage lies in pro-

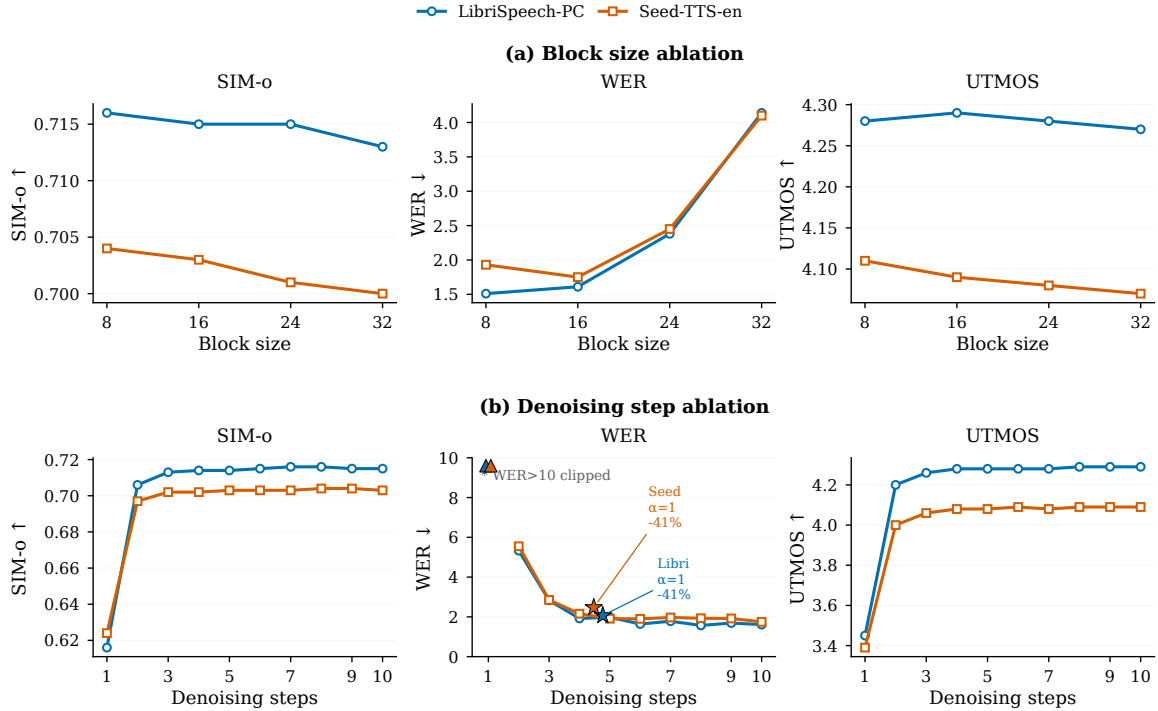


Figure 2: Inference-time ablations on LibriSpeech-PC (blue, circles) and Seed-TTS test-en (orange, squares). **(a) Block size D** and **(b) denoising step budget K** (fixed-step, $\alpha=0$). Triangles: WER off the display range ($K \leq 2$). Stars: adaptive early decoding ($\alpha=1$), reaching plateau-level WER at ~ 4.6 steps/block. Trends discussed in Section 3.6.

viding the calibrated confidence signal that enables adaptive early decoding.

4 Conclusion

We presented Chatterbox-Flash, a block-diffusion zero-shot TTS model obtained by fine-tuning a pretrained autoregressive decoder into a parallel masked decoder while preserving block-by-block streaming. Our key contributions are an inference-time prior-calibrated scoring scheme, which suppresses the dominant-token bias of discrete speech codecs and provides a well-calibrated confidence signal; an early-decoding schedule that adaptively terminates iteration; and a streaming-compatible inference engine that combines the two. On standard zero-shot TTS benchmarks, it matches strong AR and NAR baselines in quality despite training on substantially less data, achieves the highest UTMOS among NAR baselines on both LibriSpeech-PC and Seed-TTS, and reduces the average denoising step count by $\sim 20\%$ at minimal quality cost. While prior calibration is largely interchangeable with a top-confidence baseline on saturated read-speech benchmarks, its value emerges in the regime the model is built for: on challenging inputs decoded under streaming, it keeps word error low

where the baseline degrades sharply. These results indicate that block diffusion with calibration-aware inference is a viable design point for production-grade streaming zero-shot TTS.

Limitations

We prioritized training stability over data-related ablations, leaving the contribution of individual data sources unisolated, and the model still collapses at substantially larger block sizes ($D \geq 128$) while OmniVoice’s full-sequence formulation remains stable; alternative recipes (Section F) reach larger D only under restricted conditions, leaving the parallelism gap with full-sequence models open. Finally, at saturated compute PMI and the TS schedule baseline yield statistically equivalent objective metrics (Table 7), so PMI’s advantage there is the calibrated confidence it provides for early decoding rather than a direct quality gain; on harder out-of-distribution inputs it does reduce WER directly, but whether this holds under tighter budgets or out-of-domain reference speakers is left to future work.

References

- Sand. ai, Hansi Teng, Hongyu Jia, Lei Sun, Lingzhi Li, Maolin Li, Mingqiu Tang, Shuai Han, Tianning Zhang, W. Q. Zhang, Weifeng Luo, Xiaoyang Kang, Yuchen Sun, Yue Cao, Yunpeng Huang, Yutong Lin, Yuxin Fang, Zewei Tao, Zheng Zhang, and 20 others. 2025. [Magi-1: Autoregressive video generation at scale](#). *Preprint*, arXiv:2505.13211.
- Philip Anastassiou, Jiawei Chen, Jitong Chen, Yuanzhe Chen, Zhuo Chen, Ziyi Chen, Jian Cong, Lelai Deng, Chuang Ding, Lu Gao, and 1 others. Seed-tts: A family of high-quality versatile speech generation models, 2024. URL <https://arxiv.org/abs/2406.02430>.
- Marianne Arriola, Naveen Venkat, Jonathan Granskog, and Anastasis Germanidis. 2025a. Adapting autoregressive vision language models for parallel diffusion decoding. <https://runwayml.com/research/autoregressive-to-diffusion-vlms>. Runway Research.
- Marianne Arriola and 1 others. 2025b. Block diffusion: Interpolating between autoregressive and diffusion language models. *arXiv preprint arXiv:2503.09573*.
- Hikaru Asano, Tadashi Kozuno, Kuniaki Saito, and Yukino Baba. 2026. Where-to-unmask: Ground-truth-guided unmasking order learning for masked diffusion language models. *arXiv preprint arXiv:2602.09501*.
- Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. 2021. Structured denoising diffusion models in discrete state-spaces. In *NeurIPS*.
- Evelina Bakhturina, Vitaly Lavrukhin, Boris Ginsburg, and Yang Zhang. 2021. [Hi-fi multi-speaker english tts dataset](#). *Preprint*, arXiv:2104.01497.
- Tiwei Bie, Maosong Cao, Xiang Cao, Bingsen Chen, Fuyuan Chen, Kun Chen, Lun Du, Daozhuo Feng, Haibo Feng, Mingliang Gong, and 1 others. 2026. Llada2.1: Speeding up text diffusion via token editing. *arXiv preprint arXiv:2602.08676*.
- Tiwei Bie, Maosong Cao, Kun Chen, Lun Du, Mingliang Gong, Zhuochen Gong, Yanmei Gu, Jiaqi Hu, Zenan Huang, Zhenzhong Lan, and 1 others. 2025. Llada2.0: Scaling up diffusion language models to 100b. *arXiv preprint arXiv:2512.15745*.
- Zalán Borsos, Raphaël Marinier, Damien Vincent, Eugene Kharitonov, Olivier Pietquin, Matt Sharifi, Dominik Roblek, Olivier Teboul, David Grangier, Marco Tagliasacchi, and 1 others. 2023. Audioldm: a language modeling approach to audio generation. *IEEE/ACM transactions on audio, speech, and language processing*, 31:2523–2533.
- Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangzhan Yu, and Furu Wei. 2022. [Wavlm: Large-scale self-supervised pre-training for full stack speech processing](#). *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518.
- Yushen Chen, Zhikang Niu, Ziyang Ma, Keqi Deng, Chunhui Wang, JianZhao JianZhao, Kai Yu, and Xie Chen. 2025. F5-tts: A fairytale that fakes fluent and faithful speech with flow matching. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6255–6271.
- Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi. 2022. High fidelity neural audio compression. *arXiv preprint arXiv:2210.13438*.
- Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. 2020. [ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification](#). In *Interspeech 2020*, pages 3830–3834.
- Zhihao Du, Changfeng Gao, Yuxuan Wang, Fan Yu, Tianyu Zhao, Hao Wang, Xiang Lv, Hui Wang, Chongjia Ni, Xian Shi, and 1 others. 2025. Cosyvoice 3: Towards in-the-wild speech generation via scaling-up and post-training. *arXiv preprint arXiv:2505.17589*.
- Zhihao Du, Yuxuan Wang, Qian Chen, Xian Shi, Xiang Lv, Tianyu Zhao, Zhifu Gao, Yexin Yang, Changfeng Gao, Hui Wang, and 1 others. 2024. Cosyvoice 2: Scalable streaming speech synthesis with large language models. *arXiv preprint arXiv:2412.10117*.
- Sefik Emre Eskimez, Xiaofei Wang, Manthan Thakker, Canrun Li, Chung-Hsien Tsai, Zhen Xiao, Hemin Yang, Zirun Zhu, Min Tang, Xu Tan, and 1 others. 2024. E2 tts: Embarrassingly easy fully non-autoregressive zero-shot tts. In *2024 IEEE spoken language technology workshop (SLT)*, pages 682–689. IEEE.
- Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, and Kaiming He. 2025. [Mean flows for one-step generative modeling](#). *Preprint*, arXiv:2505.13447.
- Haorui He, Zengqiang Shang, Chaoren Wang, Xuyuan Li, Yicheng Gu, Hua Hua, Liwei Liu, Chen Yang, Jiaqi Li, Peiyang Shi, Yuancheng Wang, Kai Chen, Pengyuan Zhang, and Zhizheng Wu. 2024. [Emilia: An extensive, multilingual, and diverse speech dataset for large-scale speech generation](#). *Preprint*, arXiv:2407.05361.
- Jonathan Ho and Tim Salimans. 2022. [Classifier-free diffusion guidance](#). *Preprint*, arXiv:2207.12598.
- Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. 2021. [Hubert: Self-supervised speech representation learning by masked prediction of hidden units](#). *Preprint*, arXiv:2106.07447.

- Hangrui Hu, Xinfu Zhu, Ting He, Dake Guo, Bin Zhang, Xiong Wang, Zhifang Guo, Ziyue Jiang, Hongkun Hao, Zishan Guo, and 1 others. 2026. Qwen3-tts technical report. *arXiv preprint arXiv:2601.15621*.
- Yuma Koizumi, Heiga Zen, Shigeki Karita, Yifan Ding, Kohei Yatabe, Nobuyuki Morioka, Michiel Bacchi-ani, Yu Zhang, Wei Han, and Ankur Bapna. 2023. *Libritts-r: A restored multi-speaker text-to-speech corpus*. *Preprint*, arXiv:2305.18802.
- Pin-Jui Ku, He Huang, Jean-Marie Lemerrier, Subham Sekhar Sahoo, Zhehuai Chen, and Ante Jukić. 2026. Discrete diffusion for generative modeling of text-aligned speech tokens. In *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 17022–17026. IEEE.
- Matthew Le, Apoorv Vyas, Bowen Shi, Brian Karrer, Leda Sari, Rashel Moritz, Mary Williamson, Vimal Manohar, Yossi Adi, Jay Mahadeokar, and 1 others. 2023. Voicebox: Text-guided multilingual universal speech generation at scale. *Advances in neural information processing systems*, 36:14005–14034.
- Shufan Li, Konstantinos Kallidromitis, Hritik Bansal, Akash Gokul, Yusuke Kato, Kazuki Kozuka, Jason Kuen, Zhe Lin, Kai-Wei Chang, and Aditya Grover. 2026. Lavida: A large diffusion language model for multimodal understanding. *Advances in Neural Information Processing Systems*, 38:105101–105134.
- Wenrui Liu, Qian Chen, Wen Wang, Yafeng Chen, Jin Xu, Zhifang Guo, Guanrou Yang, Weiqin Li, Xiaoda Yang, Tao Jin, Minghui Fang, Jialong Zuo, Jionghao Bai, and Zemin Liu. 2025. *Speech token prediction via compressed-to-fine language modeling for speech generation*. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*. ArXiv:2505.24496.
- Aaron Lou, Chenlin Meng, and Stefano Ermon. 2024. Discrete diffusion modeling by estimating the ratios of the data distribution. In *ICML*.
- Ruskin Raj Manku, Yuzhi Tang, Xingjian Shi, Mu Li, and Alexander Smola. 2026. Emergenttts-eval: Evaluating tts models on complex prosodic, expressiveness, and linguistic challenges using model-as-a-judge. *Advances in Neural Information Processing Systems*, 38.
- Aleksandr Meister, Matvei Novikov, Nikolay Karpov, Evelina Bakhturina, Vitaly Lavrukhin, and Boris Ginsburg. 2023. *Librispeech-pc: Benchmark for evaluation of punctuation and capitalization capabilities of end-to-end asr models*. *Preprint*, arXiv:2310.02943.
- Tu Anh Nguyen, Wei-Ning Hsu, Antony D’Avirro, Bowen Shi, Itai Gat, Maryam Fazel-Zarani, Tal Re-merz, Jade Copet, Gabriel Synnaeve, Michael Hassid, Felix Kreuk, Yossi Adi, and Emmanuel Dupoux. 2023. *Expresso: A benchmark and analysis of discrete expressive speech resynthesis*. *Preprint*, arXiv:2308.05725.
- Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. 2026. Large language diffusion models. *Advances in Neural Information Processing Systems*, 38:50608–50646.
- Titouan Parcollet, Yuan Tseng, Shucong Zhang, and Rogier van Dalen. 2025. *Loquacious set: 25,000 hours of transcribed and diverse english speech recognition data for research and commercial use*. *Preprint*, arXiv:2505.21578.
- Vineel Pratap, Qiantong Xu, Anuroop Sriram, Gabriel Synnaeve, and Ronan Collobert. 2020. Mls: A large-scale multilingual dataset for speech research. In *Proc. Interspeech 2020*, pages 2757–2761.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2022. *Robust speech recognition via large-scale weak supervision*. *Preprint*, arXiv:2212.04356.
- Resemble AI. 2025. Chatterbox-TTS. <https://github.com/resemble-ai/chatterbox>. GitHub repository.
- Julius Richter, Yi-Chiao Wu, Steven Krenn, Simon Welker, Bunlong Lay, Shinji Watanabe, Alexander Richard, and Timo Gerkmann. 2024. *Ears: An anechoic fullband speech dataset benchmarked for speech enhancement and dereverberation*. *Preprint*, arXiv:2406.06185.
- Junhao Ruan, Bei Li, Yongjing Yin, Pengcheng Huang, Xin Chen, Jingang Wang, Xunliang Cai, Tong Xiao, and JingBo Zhu. 2026. *Causal autoregressive diffusion language model*. *Preprint*, arXiv:2601.22031.
- Takaaki Saeki, Detai Xin, Wataru Nakata, Tomoki Koriyama, Shinnosuke Takamichi, and Hiroshi Saruwatari. 2022. *Utmos: Utkyo-sarulab system for voicemos challenge 2022*. *Preprint*, arXiv:2204.02152.
- Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T. Chiu, Alexander M. Rush, and Volodymyr Kuleshov. 2024. Simple and effective masked diffusion language models. In *NeurIPS*.
- Kai Shen, Zeqian Ju, Xu Tan, Eric Liu, Yichong Leng, Lei He, Tao Qin, Jiang Bian, and 1 others. 2024. Naturalspeech 2: Latent diffusion models are natural and zero-shot speech and singing synthesizers. In *International conference on learning representations*, volume 2024, pages 698–722.
- Yingte Shu, Yuchuan Tian, Chao Xu, Yunhe Wang, and Hanting Chen. 2026. Deferred commitment decoding for diffusion language models with confidence-aware sliding windows. *arXiv preprint arXiv:2601.02076*.
- Amitay Sicherman and Yossi Adi. 2023. Analysing discrete self supervised speech representation for spoken language modeling. In *ICASSP 2023 - 2023 IEEE*

- International Conference on Acoustics, Speech and Signal Processing.*
- Yakun Song, Xiaobin Zhuang, Jiawei Chen, Zhikang Niu, Guanrou Yang, Chenpeng Du, Dongya Jia, Zhuo Chen, Yuping Wang, Yuxuan Wang, and 1 others. 2025. Distar: Diffusion over a scalable token autoregressive representation for speech generation. *arXiv preprint arXiv:2510.12210*.
- Li Wan, Quan Wang, Alan Papir, and Ignacio Lopez Moreno. 2020. [Generalized end-to-end loss for speaker verification](#). *Preprint*, arXiv:1710.10467.
- Chengyi Wang, Sanyuan Chen, Yu Wu, Ziqiang Zhang, Long Zhou, Shujie Liu, Zhuo Chen, Yanqing Liu, Huaming Wang, Jinyu Li, Lei He, Sheng Zhao, and Furu Wei. 2023. [Neural codec language models are zero-shot text to speech synthesizers](#). *Preprint*, arXiv:2301.02111.
- Wenbin Wang, Yang Song, and Sanjay Jha. 2024. [Globe: A high-quality english corpus with global accents for zero-shot speaker adaptive text-to-speech](#). *Preprint*, arXiv:2406.14875.
- Yuancheng Wang, Haoyue Zhan, Liwei Liu, Ruihong Zeng, Haotian Guo, Jiachen Zheng, Qiang Zhang, Xueyao Zhang, Shunsi Zhang, and Zhizheng Wu. 2025. Maskgct: Zero-shot text-to-speech with masked generative codec transformer. In *International Conference on Learning Representations*, volume 2025, pages 47127–47150.
- Chengyue Wu, Hao Zhang, Shuchen Xue, Shizhe Diao, Yonggan Fu, Zhijian Liu, Pavlo Molchanov, Ping Luo, Song Han, and Enze Xie. 2025a. [Fast-dllm v2: Efficient block-diffusion llm](#). *Preprint*, arXiv:2509.26328.
- Chengyue Wu and 1 others. 2025b. Fast-dllm: Training-free acceleration of diffusion llm by enabling kv cache and parallel decoding. *arXiv preprint arXiv:2505.22618*.
- Yifan Yang, Bing Han, Hui Wang, Long Zhou, Wei Wang, Mingyu Cui, Xu Tan, and Xie Chen. 2026. Measuring prosody diversity in zero-shot tts: A new metric, benchmark, and exploration. In *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 17252–17256. IEEE.
- Jiacheng Ye, Zhihui Xie, Lin Zheng, Jiahui Gao, Zirui Wu, Xin Jiang, Zhenguo Li, and Lingpeng Kong. 2025a. Dream 7b: Diffusion large language models. *arXiv preprint arXiv:2508.15487*.
- Zihao Ye, Lequn Chen, Ruihang Lai, Wuwei Lin, Yineng Zhang, Stephanie Wang, Tianqi Chen, Baris Kasikci, Vinod Grover, Arvind Krishnamurthy, and Luis Ceze. 2025b. [Flashinfer: Efficient and customizable attention engine for llm inference serving](#). *Preprint*, arXiv:2501.01005.
- Neil Zeghidour, Alejandro Luebs, Ahmed Omran, Jan Skoglund, and Marco Tagliasacchi. 2021. Soundstream: An end-to-end neural audio codec. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 30:495–507.
- Tao Zewei and Huang Yunpeng. 2025. Magiattention: A distributed attention towards linear scalability for ultra-long context, heterogeneous mask training. <https://github.com/SandAI-org/MagiAttention/>.
- Siyi Zhou, Yiquan Zhou, Yi He, Xun Zhou, Jinchao Wang, Wei Deng, and Jingchen Shu. 2026. Indextts2: A breakthrough in emotionally expressive and duration-controlled auto-regressive zero-shot text-to-speech. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 35139–35148.
- Yixuan Zhou, Guoyang Zeng, Xin Liu, Xiang Li, Renjie Yu, Ziyang Wang, Runchuan Ye, Weiyue Sun, Jiancheng Gui, Kehan Li, and 1 others. 2025. Voxcpm: Tokenizer-free tts for context-aware speech generation and true-to-life voice cloning. *arXiv preprint arXiv:2509.24650*.
- Han Zhu, Wei Kang, Zengwei Yao, Liyong Guo, Fangjun Kuang, Zhaoqing Li, Weiwei Zhuang, Long Lin, and Daniel Povey. 2025. Zipvoice: Fast and high-quality zero-shot text-to-speech with flow matching. *arXiv preprint arXiv:2506.13053*.
- Han Zhu, Lingxuan Ye, Wei Kang, Zengwei Yao, Liyong Guo, Fangjun Kuang, Zhifeng Han, Weiwei Zhuang, Long Lin, and Daniel Povey. 2026. Omnivoice: Towards omnilingual zero-shot text-to-speech with diffusion language models. *arXiv preprint arXiv:2604.00688*.

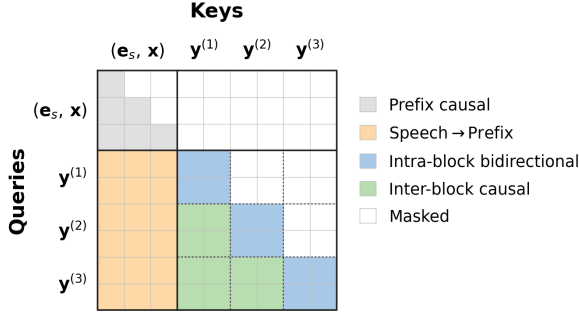


Figure 3: Hybrid block attention mask at $D = 2$. A colored cell at row i , column j indicates that query position i attends to key position j . The conditioning prefix (e_s, x) uses causal attention (gray); speech tokens attend to the entire prefix (orange); within each speech block $y^{(b)}$, attention is bidirectional (blue); across blocks, attention is causal (green). Future blocks are never visible to past blocks, enabling block-by-block streaming.

A Attention Kernel Implementation

The hybrid attention mask described in Section 2.2 is implemented using either (i) PyTorch’s native `flex_attention` with a custom mask function, or (ii) MagiAttention’s Flex-Flash-Attention (FFA) (ai et al., 2025; Zewei and Yunpeng, 2025)² kernel for higher throughput on long sequences.

Range Encoding for FFA FFA expresses sparse attention as a list of rectangular query/key index ranges with per-range attention types (FULL or CAUSAL). Letting $L_{\text{pre}} = |e_s| + |x|$ denote the prefix length, N the speech-stream length, and D the block size, our hybrid mask decomposes into four rectangle groups:

1. **Prefix Self-Attention** (CAUSAL): $q, k \in [0, L_{\text{pre}})$.
2. **Speech-to-Prefix** (FULL): $q \in [L_{\text{pre}}, L_{\text{pre}} + N), k \in [0, L_{\text{pre}})$.
3. **Intra-Block Bidirectional** (FULL): for each block b , $q, k \in [L_{\text{pre}} + bD, L_{\text{pre}} + (b+1)D)$.
4. **Inter-Block Left-Context** (FULL): for $b \geq 1$, q in block b , k in blocks $0, \dots, b-1$.

Equivalence The boolean mask induced by the FFA range union is identical to the reference mask used by `flex_attention`; both backends produce numerically equivalent outputs up to bf16 noise.

²<https://github.com/SandAI-org/MagiAttention>

B Training Dataset Details

We train on approximately 70k hours of English speech (44M utterances, 528k speakers), compiled from publicly available corpora—spanning large-scale read speech (Pratap et al., 2020; He et al., 2024; Parcollet et al., 2025; Koizumi et al., 2023; Bakhturina et al., 2021), expressive and anechoic speech (Nguyen et al., 2023; Richter et al., 2024), and accented English (Wang et al., 2024)—together with privately collected audio-book, conversational, and short-form utterances (names, numbers, time expressions). Table 6 lists the datasets used for training. The total training set contains 43.8M utterances (~ 70 k hours) from 528k speakers, distributed across public and privately collected sources. The privately collected portion broadens coverage of conversational, expressive, and short-form utterances beyond standard read-speech corpora.

C Additional Ablations

This appendix reports additional ablations supplementing Section 3.6: sampling and position temperatures (Section C.1) and the CFG scale w (Section C.2). All sweeps use the canonical configuration of Section 3.1 except for the swept parameter.

C.1 Sampling and Position Temperatures

Sampling Temperature Table 5 sweeps the speech-token sampling temperature T at the canonical configuration ($D = 16$, $K = 10$, $\alpha = 0$, $w = 1.0$, $\beta = 5$). $T = 0.2$ minimizes mean WER on both benchmarks (1.61 on LibriSpeech-PC, 1.75 on Seed-TTS) and is our default; increasing T progressively degrades WER (+0.4 at $T = 1.0$) while leaving SIM-o and UTMOS within noise.

Position Temperature and Decoding Method

For the position temperature β (Gumbel perturbation on the prior-calibrated scores; Section 2.3.3), we compared $\beta = 0$ (deterministic top- n_k) against $\beta = 5$ across both decoding methods (TS schedule and PMI) and step budgets $K \in \{2, 5, 8\}$ (Table 7). This sweep simultaneously serves as the head-to-head comparison between the TS schedule baseline and PMI referenced in Section 3.6. $\beta = 5$ consistently lowers WER over $\beta = 0$ on both benchmarks across all settings, with the effect most pronounced at low K (e.g., at $K = 2$: PMI WER drops from 11.82 to 5.17 on LibriSpeech-PC, 13.98 to 5.37 on Seed-TTS); the same trend holds

T	LibriSpeech-PC test-clean			Seed-TTS test-en		
	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
0.2 (default)	0.715	1.61	4.29	0.703	1.75	4.09
0.4	0.715	1.64	4.29	0.704	1.82	4.08
0.6	0.714	1.75	4.28	0.702	1.93	4.07
0.8	0.713	1.75	4.26	0.702	2.05	4.05
1.0	0.714	2.01	4.24	0.702	2.15	4.02

Table 5: Sampling temperature T sweep at $\beta = 5$, others at canonical default.

Dataset	# Samples
<i>Public</i>	
MLS-English (Pratap et al., 2020)	10.8M
Emilia (en, part 1) (He et al., 2024)	9.1M
Loquacious (Parcollet et al., 2025)	3.9M
GLOBE (Wang et al., 2024)	582K
LibriTTS-R (Koizumi et al., 2023)	375K
HiFi-TTS (Bakhturina et al., 2021)	324K
EARS (Richter et al., 2024)	12K
Expresso (Nguyen et al., 2023)	12K
<i>Privately collected</i>	
Audiobook	17.7M
Podcasts	726K
IVR (VC-augmented)	445K
Short-form utterances [†]	292K
Conversational	50K
Stylized speech	62K
Total	43.8M (~70k hours)

Table 6: Training dataset composition. Sample counts are reported after preprocessing and integrity filtering. [†]Short-form includes phone numbers, names, time expressions, and expressive single-word utterances.

for the TS schedule. Comparing the TS schedule baseline and PMI at $\beta = 5$, the two methods track each other closely across all step budgets—PMI is slightly ahead at $K = 5, 8$ on LibriSpeech-PC and at $K = 2$ on both benchmarks, while the TS schedule is slightly ahead at $K = 5$ on Seed-TTS—suggesting that, on these read-speech benchmarks where WER saturates near 1.7–2.0, the choice of decoding method contributes only marginally to quality. PMI’s empirical advantage manifests in two regimes beyond this saturated setting: when early decoding is applied (e.g., $\alpha = 0.5$ in Table 1), where the calibrated confidence signal enables the per-step quantile threshold of Equation (9) to terminate blocks early without quality regression; and on harder, out-of-distribution inputs (Table 3), where prior calibration directly reduces WER. SIM-o and UTMOS also improve modestly with $\beta = 5$. We adopt $\beta = 5$ throughout.

C.2 Classifier-Free Guidance Scale

Table 8 sweeps the CFG scale w at the canonical configuration ($D = 16$, $K = 10$, $\alpha = 0$, $T = 0.2$, $\beta = 5$). The two benchmarks favor slightly different operating points: LibriSpeech-PC attains its lowest WER at $w = 0.5$ (1.52), while Seed-TTS test-en attains its lowest at $w = 1.5$ (1.77). $w = 1.0$ provides the most balanced trade-off across both—1.63 / 1.79 WER with the highest Seed-TTS SIM-o (0.705)—and is our default. Larger w (≥ 1.5) marginally trades intelligibility against SIM-o and UTMOS on Seed-TTS.

D Streaming Efficiency

This appendix provides the full streaming efficiency numbers summarized in Section 3.2.

Comparability Caveat The Qwen3-TTS (Hu et al., 2026) numbers reported below are taken from its own technical report, not measured by us. The report does not fully specify the GPU, optimization, batching, or concurrency level used, and the systems also differ in model size and token rate (12 Hz vs. 25 Hz variants). We therefore do *not* treat the two sets of numbers as a controlled, like-for-like comparison, and we avoid drawing head-to-head speedup conclusions from them. We include the Qwen3-TTS numbers only as a loose reference point—the best publicly available figures for a comparable streaming AR TTS system—to situate our measurements in context. All Chatterbox-Flash numbers are measured by us under the protocol below.

Measurement Protocol Both metrics are computed on 50 utterances at concurrency 1, measured up to the moment the first audio packet is emitted by the server. *Time to first packet* (TTFP) is the wall-clock time from receiving the request to emitting the first audio packet; *real-time factor* (RTF) is the ratio of generation wall-clock time to synthesized audio duration. All Chatterbox-Flash num-

K	Method	β	LibriSpeech-PC test-clean			Seed-TTS test-en		
			SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
2	TS schedule	0	0.679	15.10	4.00	0.676	15.40	3.81
	TS schedule	5	0.697	8.74	4.13	0.690	9.13	3.93
	PMI	0	0.690	11.82	4.08	0.682	13.98	3.86
	PMI	5	0.707	5.17	4.21	0.697	5.37	4.01
5	TS schedule	0	0.714	2.52	4.26	0.702	2.50	4.08
	TS schedule	5	0.714	1.91	4.28	0.702	2.13	4.09
	PMI	0	0.715	2.36	4.27	0.703	2.46	4.07
	PMI	5	0.713	1.96	4.27	0.703	2.01	4.08
8	TS schedule	0	0.714	1.87	4.28	0.703	2.07	4.09
	TS schedule	5	0.714	1.69	4.29	0.703	1.97	4.09
	PMI	0	0.716	1.82	4.28	0.703	2.25	4.09
	PMI	5	0.717	1.67	4.29	0.704	1.96	4.09

Table 7: Position temperature β ablation at $D = 16$, $\tau = 0.5$, $w = 1.0$, $T = 0.2$. $\beta = 5$ (Gumbel-perturbed selection) consistently outperforms $\beta = 0$ (deterministic top- n_k) across both decoding methods (TS schedule, PMI) and all step budgets, with the effect most pronounced at low K . Best WER per K is in **bold**.

w	LibriSpeech-PC test-clean			Seed-TTS test-en		
	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
0.5	0.715	1.52	4.30	0.700	1.94	4.12
1.0 (default)	0.715	1.63	4.29	0.705	1.79	4.08
1.5	0.715	1.71	4.27	0.703	1.77	4.05
2.0	0.714	1.80	4.26	0.702	1.94	4.02

Table 8: CFG scale w sweep at the canonical configuration. $w = 1.0$ is the default used in our main results.

Model / Config	TTFP (ms) $\downarrow$	RTF $\downarrow$
<i>Qwen3-TTS (Hu et al., 2026) (AR, reported)</i>		
25 Hz, 1.7B	150	0.253
25 Hz, 0.6B	138	0.234
12 Hz, 1.7B	101	0.313
12 Hz, 0.6B	97	0.288
<i>Chatterbox-Flash (ours)</i>		
$D = 16$, $\alpha = 0.5$ (default)	118	0.107
$D = 16$, $\alpha = 0.75$	106	0.091
$D = 24$, $\alpha = 0.5$	119	0.100
$D = 24$, $\alpha = 0.75$	105	0.084
$D = 32$, $\alpha = 0.5$	115	0.090
$D = 32$, $\alpha = 0.75$	103	0.076

Table 9: Streaming efficiency at concurrency 1 over 50 utterances. **TTFP**: request-to-first-packet wall-clock time; **RTF**: generation time / audio duration. Chatterbox-Flash is 25 Hz, 0.5B, measured by us on H100; best values in **bold**. Qwen3-TTS numbers are reproduced from its technical report under its own, not fully specified, setup—not re-measured by us—and serve only as a loose reference, not a controlled comparison (see Comparability Caveat). Differences in GPU, precision, batching, model size, and token rate mean the two blocks are not directly comparable.

bers are measured on NVIDIA H100 GPUs using FlashInfer attention kernels and paged key-value

cache management (Section E).

Results Under the efficiency-oriented configuration ($D = 16$, $\alpha = 0.5$), Chatterbox-Flash attains TTFP 118 ms and RTF 0.107 (Table 9). Its measured TTFP falls in a similar range to the Qwen3-TTS figures (97–150 ms across the four reported variants), and its RTF is numerically lower than all of them (0.234–0.313 as reported), corresponding to roughly $9\times$ real-time synthesis at concurrency 1 in our setup. More aggressive settings reduce these numbers further: at $D = 32$, $\alpha = 0.75$, Chatterbox-Flash records TTFP 103 ms and RTF 0.076 ($\sim 13\times$ real time). Because the two systems were measured under different and not fully specified conditions, we report these values side by side to indicate that Chatterbox-Flash operates in a competitive latency-throughput regime, rather than to claim a controlled speedup over Qwen3-TTS.

D.1 Streaming Server Quality

Table 10 reports objective quality measured on the production streaming server used for the latency numbers above (Table 9), i.e. with the chunk-wise vocoder and early-emit block serving enabled. This differs from the controlled comparison of Table 2,

Config	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
$D = 16, \alpha = 0.5$ (default)	0.688	2.03	4.07
$D = 16, \alpha = 0.75$	0.687	2.27	4.06
$D = 24, \alpha = 0.5$	0.687	2.63	4.07
$D = 24, \alpha = 0.75$	0.688	2.74	4.07
$D = 32, \alpha = 0.5$	0.686	3.86	4.06
$D = 32, \alpha = 0.75$	0.685	3.70	4.05
<i>Offline reference (Table 2)</i>			
$D = 16, \alpha = 0.5$	0.713	1.67	4.28

Table 10: Production streaming-server quality on LibriSpeech-PC test-clean, under the same configurations as Table 9 (chunk vocoder + early-emit enabled). Unlike the decoder-only comparison in Table 2, these numbers include server-level sampling optimizations and vary block size, so a modest quality drop versus offline appears, growing in WER at larger D .

which isolates the decoder at a fixed configuration: the production server additionally varies block size and applies sampling-level optimizations, so its numbers reflect end-to-end deployment behavior and can differ from the decoder-only measurements. At the default $D = 16, \alpha = 0.5$, the server attains SIM-o 0.688, WER 2.03, and UTMOS 4.07 on LibriSpeech-PC test-clean. Across configurations, SIM-o and UTMOS stay stable while WER grows with block size, consistent with the block-size ablation (Figure 2); this is the main quality cost of the most aggressive latency settings.

Vocoder Chunk Schedule The chunk-wise vocoder operates with a progressively widening chunk schedule. The first chunk has a duration of 0.46 s (~ 12 tokens at the 25 Hz codec rate), with each subsequent chunk grown by a factor of 5.0, capped at 6.0 s (150 tokens). A small initial chunk reduces TTFP, since the vocoder can emit the first audio packet as soon as a short window of speech tokens is available; subsequent larger chunks amortize vocoder overhead during sustained synthesis. The chunk duration can be tuned per deployment scenario, with shorter initial chunks reducing TTFP at the cost of slightly higher steady-state cost.

Early-Emit Block Serving We further reduce streaming latency through an *early-emit* optimization at the block boundary. During block decoding, masked positions within a block are unmasked progressively, and the early-decoding schedule (see [Early Decoding](#) paragraph) typically commits positions in a non-contiguous order. Whenever the committed positions form a contiguous left-aligned prefix of the block—that is, the remaining masked po-

sitions occupy only the right portion—we emit the prefix tokens to the vocoder immediately, without waiting for the entire block to finish decoding. This overlaps vocoder decoding of the early-committed tokens with the remaining denoising steps for the same block, yielding additional latency reduction while preserving the block-by-block streaming abstraction.

E Implementation Details

The inference engine is built on FlashInfer, with three customizations specific to block-diffusion TTS that differ from standard LLM serving.

Hybrid Causal/Non-Causal Attention Standard LLM serving uses causal attention throughout. Our inference loop instead interleaves two attention patterns within a single decoding step: the conditioning prefix is encoded with causal attention and cached once, while the current block is decoded with non-causal attention so that all masked positions in the block attend bidirectionally to each other (Section 2.3.1). The two patterns share the same paged key-value buffer and differ only in the causal flag passed to the attention kernel.

Frozen Prefix, Growing Block Cache The prefix forward is invoked once per generation and writes the prefix key-value entries into the paged buffer. Subsequent block forwards reuse this frozen prefix cache, append the current block’s key-value entries, and never recompute prefix attention. This realizes the two-cache abstraction described in Section 2.3.1.

Cache Snapshot for CFG Classifier-free guidance (Section 2.3.4) requires both a conditional and an unconditional forward at each block step. We snapshot the paged cache before the unconditional pass and restore it afterwards, so the conditional state is preserved without recomputing the prefix.

CUDA Graph Replay Because the block size D is fixed at inference time, the per-step forward—attention, feed-forward, and the speech head—is captured as a single CUDA graph and replayed for every step of every block. This removes the per-launch overhead that otherwise dominates short-sequence, small-batch inference.

F Block-Size Scaling Explorations

This appendix summarizes alternative block-diffusion configurations we explored when attempt-

ing to scale beyond the block sizes used in our main results. None of these alternatives delivered both stable training at large block sizes and the prosodic quality required for production deployment, but they characterize where the difficulty lies and motivate future work.

Fully Causal Block Formulation (CARD-style)

Following the causal autoregressive diffusion formulation of CARD (Ruan et al., 2026), we trained a variant in which all attention within and across blocks is strictly causal, removing the intra-block bidirectional context of our default formulation. At small block sizes ($D \leq 4$), this variant produces intelligible speech and offers the fastest inference among the configurations we tested, since a single forward pass per block is sufficient and no iterative refinement is required. Starting from $D = 5$, however, generated speech begins to exhibit prosodic collapse and over-smoothing, with reduced pitch and energy variation and noticeably flatter intonation. The fully causal training signal appears to be too weak to support meaningful parallelism inside the block, suggesting that the bidirectional intra-block context used by our main formulation is necessary for prosodic fidelity in TTS.

Block-Size Annealing with Self-Distillation Inspired by A2D-VL (Arriola et al., 2025a), which gradually increases the diffusion prediction window during fine-tuning, we experimented with a data-free self-distillation recipe that anneals the block size starting from $D = 1$ (purely autoregressive) and progressively widening the block. This approach was able to maintain coherent speech up to roughly $D = 8$. Beyond this point, the model’s per-position confidence collapses to a narrow range, and sampling becomes highly sensitive to temperature and CFG scale—small perturbations in the sampling configuration produce qualitatively different outputs. Extending the schedule to larger block sizes did not produce a stable regime within our compute budget.

Discussion Across these explorations, two distinct failure modes recur: (i) loss of prosodic variability when the training signal does not enforce bidirectional intra-block context, and (ii) collapsed confidence and high sampling sensitivity when block size grows faster than the model’s denoising capability adapts. Our default block-diffusion formulation with bidirectional intra-block attention and a moderate block size offered the most sta-

ble trade-off among the recipes we tested, but a principled approach to scaling block size while preserving both prosody and confidence calibration remains an open question.

G Streaming Capability of Baselines

Our main results compare Chatterbox-Flash against NAR baselines that, to our knowledge, are not designed for streaming inference. In particular, OmniVoice (Zhu et al., 2026) is formulated as a full-sequence masked diffusion model whose architecture does not natively support streaming. The authors have indicated³ that a form of *pseudo-streaming* can nonetheless be obtained by splitting the input text into smaller chunks and processing them sequentially.

We did not evaluate this configuration in the present work for two reasons. First, chunked sampling deviates from the model’s training-time assumption of full-sequence bidirectional context, and the resulting quality–latency trade-off would likely depend on chunking strategy, overlap, and re-tuning of the denoising schedule. Second, any number reported under such an adaptation would reflect our specific reimplementation rather than the original system, making a fair comparison difficult.

We view a controlled study of pseudo-streaming OmniVoice—and, more broadly, of how full-sequence diffusion models can be retrofitted for streaming—as a meaningful direction for future work, as it would help disentangle whether the streaming advantage observed in our setting stems from the block-diffusion formulation itself or from training-time alignment with the streaming inference pattern.

³<https://github.com/k2-fsa/OmniVoice/issues/6>